

VARUN SREENIVASAN

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EDUCATION

M.S. Computer Science, University of Wisconsin-Madison (Dec 2021)

G.P.A: **4.00/4.00**

B.S. Computer Science (Minor: Mathematics), University of Wisconsin-Madison (May 2020)

G.P.A: **3.90/4.00** "Distinction in the Major", and Dean's List

RELEVANT COURSEWORK

Artificial Intelligence, Machine Learning, Computer Vision, Data Science, Computer Networks, Mobile & Wireless Networks, Discrete Mathematics, Cryptography, Combinatorics, Linear Algebra & Differential Equations, Operating Systems, Algorithms, Data Structures.

EXPERIENCE

DataChat, Inc.

Senior Software Engineer

Madison, WI

Feb 2022 - Present

- Received consistently high performance ratings across quarterly assessments.
- Proposed and implemented the adoption of the [Artillery](#) load testing tool to evaluate the performance and scalability of DataChat's SaaS application. The testing helped uncover system bottlenecks that impeded application performance under load.
- Spearheaded the push towards CI/CD to streamline the release management process involving upgrades of Kubernetes hosted environments. This effort helped increase upgrade cadence and reduce manual effort.
- Led a team of developers to overhaul the Vector Database infrastructure, enabling support for [VertexAI Vector Search](#) and [Milvus](#), which were critical providers to meet business demands.
- Spearheaded design and implementation of backend solutions to expand product capabilities within the ML & GenAI space.
- Enhanced the scalability and efficiency of the GenAI platform by adding Kubernetes' Horizontal Pod Autoscaling (HPA) for all GenAI services and optimizing CPU and RAM usage. The improvements led to a sizable reduction in Kubernetes related cloud costs.
- Accelerated LLM evaluation by setting up an LLM evaluation framework, enabling faster adoption of LLMs across various use cases.
- Led the research into the feasibility of hosting DataChat's custom LLMs across various cloud providers and platforms.

National Science Foundation – IRIS HEP

[Fellow](#) – Graph Methods for Particle Tracking (High Luminosity Large Hadron Collider)

Berkeley, CA

May 2021 - August 2021

- Developed an embedding pipeline to find a distance metric between 3D hit measurements pairs, optimizing particle differentiation.
- Accelerated the nearest neighbors search by replacing Facebook's Faiss with Fixed Radius Nearest Neighbors (FRNN) on CUDA.
- Optimized a PyTorch-based embedding model with 99% efficiency and 1% purity through hyperparameter scanning.

Citrine Informatics

[NextGen-Fellow](#) – Computational Materials Science

Redwood City, CA

May 2018 - August 2018

- Multi-university research project: Competitively selected, successfully completed bootcamp & workshop at Stanford University.
- Employed a Keras implementation of RetinaNet object detection model to identify defects in metals.
- Developed evaluation pipeline to determine recall and precision metrics. Obtained a model with 85% precision and 68% recall on the test set through hyperparameter optimization. Performed analysis to determine the reason for high false negative rate.
- Presented the results at NextGen Research Symposium in Golden, CO and co-authored a [paper](#).

SOFTWARE PROJECTS

Autonomous RC Car

Helped develop a proof of concept for an open source, affordable, and performant autonomous vehicle testbed by training a PyTorch based SSD-MobileNet object detection model on a custom dataset to do live detection of traffic signs such as speed limits and stop signs. The co-authored [paper](#) was accepted and presented at the ACM MobiArch 2022 Conference.

Business Success/Viability Forecast

Developed multiple ML models using the Yelp dataset to predict whether businesses will survive the impacts of COVID-19. Performed feature engineering to obtain a final parsed dataset, created a training and validation pipeline to guide parameter selection, evaluated models on unseen test data with multiple metrics, and determined the vital features using the Permutation Importance algorithm.

LICENSES & CERTIFICATIONS

Certified Solutions Architect Associate, Certified Machine Learning Speciality, Certified Cloud Practitioner – Amazon Web Services

LANGUAGES & TECHNOLOGY SKILLSETS

Python, Java, Git, Docker, Kubernetes, AWS, GCP, Azure, CircleCI, Terraform, Helm, VS Code, Jupyter Notebook, Unix, MacOS

PUBLICATIONS

- [uCity: a miniaturized autonomous vehicle testbed](#). In Proceedings of the 17th ACM Workshop on Mobility in the Evolving Internet Architecture (MobiArch'22). Association for Computing Machinery, New York, NY, USA.
- [Multi defect detection and analysis of electron microscopy images with deep learning](#), Computational Materials Science

